SQL ASSIGNMENT 8

Subject: Database Management Systems (SQL) **Topic:** Table Joins (INNER, LEFT, RIGHT JOIN) **Status:** Completed

QUESTION 1

Create tables `restaurants` and `dishes`. Insert at least 3 restaurants and 2–3 dishes per restaurant (including sample data with unlinked records for join testing).

```
CREATE TABLE restaurants ( id INTEGER PRIMARY KEY, name VARCHAR(100), city VARCHAR(50) );
CREATE TABLE dishes ( id INTEGER PRIMARY KEY, restaurant_id INTEGER, dish_name VARCHAR(100), price
DECIMAL(10, 2) ); INSERT INTO restaurants (id, name, city) VALUES (1, 'Swagat Restaurant',
'Ahmedabad'), (2, 'Swadisht Bites', 'Mumbai'), (3, 'Urban Cafe', 'Bengaluru'), (4, 'Empty Diner',
'Delhi'); -- Restaurant with no dishes INSERT INTO dishes (id, restaurant_id, dish_name, price)
VALUES (101, 1, 'Paneer Butter Masala', 280.00), (102, 1, 'Butter Naan', 45.00), (103, 2, 'Pav
Bhaji', 150.00), (104, 2, 'Vada Pav', 40.00), (105, 3, 'Cold Coffee', 120.00), (106, 99, 'Orphan
Dish', 99.00); -- Dish with invalid restaurant_id
```

QUESTION 2

Write an SQL `INNER JOIN` query to display each dish along with its restaurant name and city.

```
SELECT d.dish_name, d.price, r.name AS restaurant_name, r.city FROM dishes d INNER JOIN
restaurants r ON d.restaurant_id = r.id;
```

QUESTION 3

Write an SQL `LEFT JOIN` query to list all restaurants and their dishes, keeping restaurants without dishes in the result.

```
SELECT r.name AS restaurant_name, r.city, d.dish_name, d.price FROM restaurants r LEFT JOIN
dishes d ON r.id = d.restaurant_id;
```

QUESTION 4

Write an SQL `RIGHT JOIN` query to display all dishes and their restaurant names, including unlinked dishes.

```
SELECT d.dish_name, d.price, r.name AS restaurant_name FROM restaurants r RIGHT JOIN dishes
d ON r.id = d.restaurant_id;
```

QUESTION 5

Spotify Scenario: Display all playlists and their songs, ensuring empty playlists still appear. Explain the chosen join type and write the query.

Join Choice: Use a `LEFT JOIN` starting from the `playlists` table. This guarantees that every single playlist is included in the output; if a playlist has no songs mapped to it, the song columns will simply return `NULL`.

```
SELECT p.playlist_name, s.song_title, s.artist FROM playlists p LEFT JOIN playlist_songs ps
ON p.playlist_id = ps.playlist_id LEFT JOIN songs s ON ps.song_id = s.song_id;
```